

EX:2	EXPLORATORY DATA ANALYSIS (EDA) – DATA IMPORT AND EXPORT
DATE:	

AIM:

To import data from different sources and export it into different file formats using Python for Exploratory Data Analysis (EDA).

ALGORITHM:

Step 1: Open Jupyter Notebook.

Step 2: Import the required Python libraries.

Step 3: Load data from different sources such as CSV, Excel, JSON, or SQL database.

Step 4: Verify that the imported data is loaded correctly.

Step 5: Perform basic data inspection and preprocessing if required.

Step 6: Export the processed data into the required file formats such as CSV, Excel, or JSON.

Step 7: Save the exported files.

Step 8: Verify that the exported files are created successfully.

PROGRAM:

```
import pandas as pd

df = pd.read_csv("indexData.csv")

print(df)
```

```
print(df.head())
```

```
print(df.info())
```

```
print(df.describe())
```

```
import pandas as pd

df = pd.read_csv("indexData.csv")

df.to_excel("indexData.xlsx", index=False)

print("Excel file created")
```

```
import pandas as pd

df_excel = pd.read_excel("indexData.xlsx")

print(df_excel.head())
```

```
import sqlite3
import pandas as pd
df = pd.read_csv("indexData.csv")
conn = sqlite3.connect("indexData.db")
df.to_sql("index_table", conn, if_exists="replace", index=False)
query = "SELECT * FROM index_table"

sql_df = pd.read_sql(query, conn)

print(sql_df.head())
```

```
import pandas as pd

url = "https://raw.githubusercontent.com/tamil0215/DVT/main/indexData.csv.zip"

df = pd.read_csv(url, compression="zip")

print(df.head())
```

```
import pandas as pd
df = pd.read_csv("indexData.csv")
df.to_csv("output.csv", index=False)
print("CSV File Exported")
```

```
import pandas as pd
df = pd.read_csv("indexData.csv")
df.to_excel("index_output.xlsx", index=False)
print("Excel File Exported")
```

```
import pandas as pd
df = pd.read_csv("indexData.csv")
df.to_json("index_output.json")
print("JSON File Exported")
```

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("indexData.csv")

index_count = df["Index"].value_counts()

index_count.plot(kind="bar")

plt.title("Number of Records for Each Stock Index")
plt.xlabel("Stock Index")
plt.ylabel("Count")

plt.show()
```

OUTPUT:

	Index	Date	Open	High	Low	Close \
0	NYA	1965-12-31	528.690002	528.690002	528.690002	528.690002
1	NYA	1966-01-03	527.210022	527.210022	527.210022	527.210022
2	NYA	1966-01-04	527.840027	527.840027	527.840027	527.840027
3	NYA	1966-01-05	531.119995	531.119995	531.119995	531.119995
4	NYA	1966-01-06	532.070007	532.070007	532.070007	532.070007
...	...	...	...	...	...	...
112452	N100	2021-05-27	1241.119995	1251.910034	1241.119995	1247.069946
112453	N100	2021-05-28	1249.469971	1259.209961	1249.030029	1256.599976
112454	N100	2021-05-31	1256.079956	1258.880005	1248.140015	1248.930054
112455	N100	2021-06-01	1254.609985	1265.660034	1254.609985	1258.579956
112456	N100	2021-06-02	1258.489990	1263.709961	1258.239990	1263.619995
...	...	...	...	...	...	...
Adj Close						
0			528.690002			0.0
1			527.210022			0.0
2			527.840027			0.0
3			531.119995			0.0
4			532.070007			0.0
...			...			...
112452			1247.069946			379696400.0
112453			1256.599976			160773400.0
112454			1248.930054			91173700.0
112455			1258.579956			155179900.0
112456			1263.619995			148465000.0

[112457 rows x 8 columns]

	Index	Date	Open	High	Low	Close \
0	NYA	1965-12-31	528.690002	528.690002	528.690002	528.690002
1	NYA	1966-01-03	527.210022	527.210022	527.210022	527.210022
2	NYA	1966-01-04	527.840027	527.840027	527.840027	527.840027
3	NYA	1966-01-05	531.119995	531.119995	531.119995	531.119995
4	NYA	1966-01-06	532.070007	532.070007	532.070007	532.070007
...	...	...	...	...	...	...
Adj Close						
0			528.690002			0.0
1			527.210022			0.0
2			527.840027			0.0
3			531.119995			0.0
4			532.070007			0.0

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 112457 entries, 0 to 112456
Data columns (total 8 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   Index       112457 non-null  object
1   Date        112457 non-null  object
2   Open        110253 non-null  float64
3   High        110253 non-null  float64
4   Low         110253 non-null  float64
5   Close       110253 non-null  float64
6   Adj Close   110253 non-null  float64
7   Volume      110253 non-null  float64
dtypes: float64(6), object(2)
memory usage: 6.9+ MB
None

```

	Open	High	Low	Close \
count	110253.000000	110253.000000	110253.000000	110253.000000
mean	7658.515222	7704.372961	7608.000422	7657.545872
std	9011.478913	9066.638548	8954.506981	9011.510444
min	54.869999	54.869999	54.869999	54.869999
25%	1855.030029	1864.510010	1843.979980	1855.060059
50%	5194.089844	5226.279785	5154.049805	5194.750000
75%	10134.299810	10207.820310	10060.349610	10134.830080
max	68775.062500	69403.750000	68516.992190	68775.062500

	Adj Close	Volume
count	110253.000000	1.102530e+05
mean	7657.351729	1.273975e+09
std	9011.608900	4.315783e+09
min	54.869999	0.000000e+00
25%	1854.179565	0.000000e+00
50%	5194.750000	4.329000e+05
75%	10134.830080	1.734314e+08
max	68775.062500	9.440374e+10

Excel file created

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0	NYA	1965-12-31	528.690002	528.690002	528.690002	528.690002
1	NYA	1966-01-03	527.210022	527.210022	527.210022	527.210022
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4	NYA	1966-01-06	532.070007	532.070007	532.070007	532.070007

	Adj Close	Volume
0	528.690002	0.0
1	527.210022	0.0
2	527.840027	0.0
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1	527.210022	0.0
2	527.840027	0.0
3	531.119995	0.0
4	532.070007	0.0

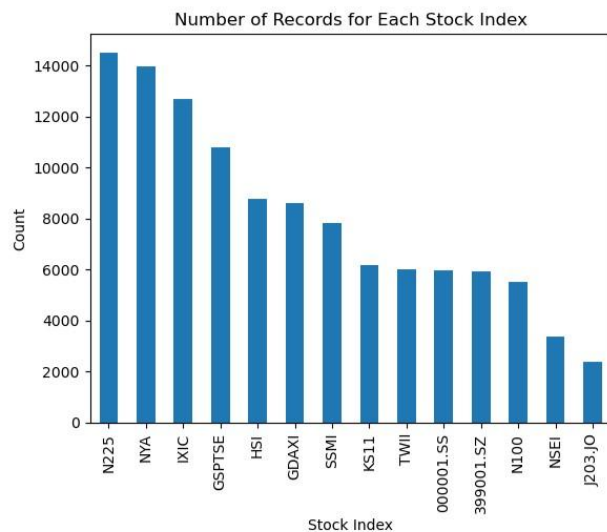
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1	527.210022	0.0
2	527.840027	0.0
3	531.119995	0.0
4	532.070007	0.0

CSV File Exported

Excel File Exported

JSON File Exported



RESULT:

Data was successfully imported from different sources and exported into different file formats using Python.